

M00845: Arginine biosynthesis, glutamate => acetylcitrulline => arginine

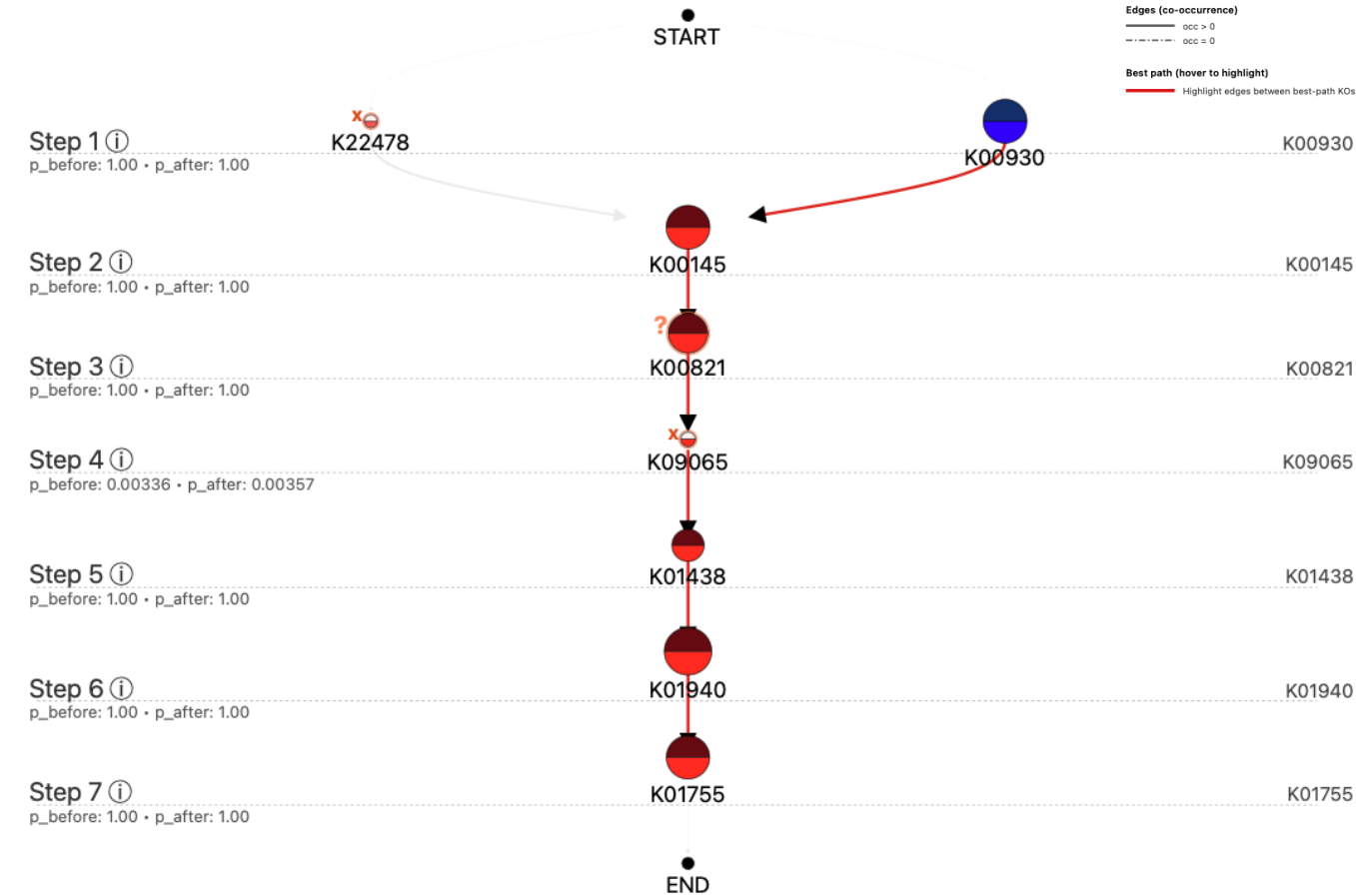
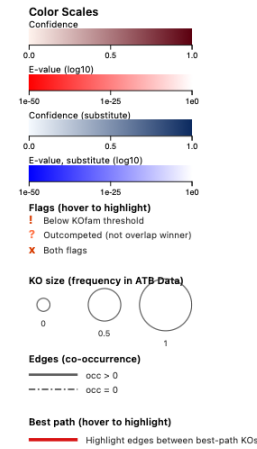
Confidence (Raw): 0.534

Confidence (After updates): 0.824

Expected steps: 7

Steps complete: 6 / 7

Arginine and proline metabolism

[Open on KEGG](#)**Node details**KO: [K22478](#)

On ORF: NC_012660.1_5881

Substitute: [K00930](#)Confidence (P_i'): 0.00106ATB Frequency (F_i): 0.0258**Calculation Inputs**

KO Score: 122 x

KOfam threshold: 516

E-value: 8.1e-36

Annotation Confidence (O_i): 1.52e-96

x Better hits for this ORF:

[K00930](#): 440Raw Confidence (P_i): 4.67e-4

Influencer shift: +5.89e-4

Node detailsKO: [K00930](#)

On ORF: NC_012660.1_5881

Raw Confidence (P_i): 1.00Confidence after propagation (P_i'): 1.00

KO Score: 440

KOfam threshold: 217

E-value: 1.90e-132

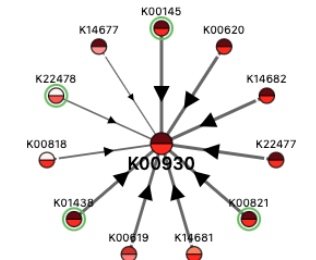
Annotation Confidence (O_i): 1.00**Influence → K00930**

E-value: 1.90e-132

• P_i : 1.00 + shift: 0 = C_i = 1.00

• Valid influencers: 11

• Modules: —



Influencing KO	Module	N_{ij}	N_j	Corr	weight	P_i	contrib
K00145	same	17230	17289	0.997	0.119	1.00	0
K00620	other	13425	13466	0.997	0.119	1.00	0
K14682	other	5895	5913	0.997	0.118	1.00	0
K22477	other	5282	5388	0.980	0.116	1.00	0
K00821	same	14144	14972	0.945	0.112	1.00	0
K14681	other	2228	2393	0.931	0.109	0.00875	0
K00619	other	10219	11248	0.909	0.108	0.193	0
K01438	same	10035	11054	0.908	0.108	1.00	0
K00818	other	34	36	0.944	0.0472	1.98e-6	0

Influence → K22478

E-value: 8.10e-36

• P_i : 4.67e-4 + shift: 5.89e-4 = C_i = 0.00106

• Valid influencers: 2

• Modules: [M00028](#), [M00845](#)

Influencing KO	Module	N_{ij}	N_j	Corr	weight	P_i	contrib
K09065	same	384	1483	0.259	0.221	0.00336	4.73e-4
K12659	other	391	392	0.997	0.779	2.35e-4	1.17e-4

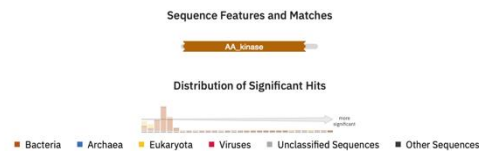
Alignment for K00930 and K22478 sequences

Query: eco:b3959 [L=258]
Description: K00930 acetylglutamate kinase [EC:2.7.2.8] | (RefSeq) argB; acetylglutamate kinase (A)
Scores for complete sequences (score includes all domains):

--- full sequence ---			--- best 1 domain ---			#dom-	Sequence	Description
E-value	score	bias	E-value	score	bias	exp N		
1.5e-22	66.6	0.9	2e-22	66.1	0.9	1.2 1	xft:PD_0293	K22478 bifunctional N-acetylglutamate synthase/K

[illegible]

Domain search for K00930 sequence



Search Details		Customise Table									
Target		Description						Species		E-value	
▼ P0A6C9		Acetylglutamate kinase						Escherichia coli		1.0e-161	
Query	Target Envelope		Target Alignment						Domain E-value		
start	end	start	end	start	end	Bias	% Identity (count)	% Similarity (count)	Bit Score	Independent	Conditional
1	258	1	258	1	258	4.4	100.0 (258)	100.0 (258)	548.0	1.2e-161	1.6e-165

Query	1	mmnpliiklgvlldseaalerfalsalvnyreshqplrvlvhggcgvdelmkglnlpvkkngkrlrvtpadqidiitgai 80									
Target	1	MMNPLIIKLGVL DSEEA LERFSA LVN YR ES HQ PLRV LVHGGCGVD ELMKGLNL PVKKNG LRVTPADQIDIITGAI 80									
PP		8*****									

Query	81	agtanktllsakkhqlaavglrlfgdgsvkvtqldeelhvglaqgspklnslenglylvvssigpytddegalmvn 160									
Target	81	AGTANKTLLS AKKHQLA AVGLRLFGDGSVKVTQLDEELHVG LAQGSPK LNSL ENG LYLVVSSIGPYTDDEGALMVN 160									
PP		*****									

Query	161	adqaatalaatlgadlllsvdsglildgpkgrlaentaakaeqlieglitdgmkvknaaldartlgrpvdisawzha 240									
Target	161	ADQAATALAATLGADLLLSVDSGLILDGPKGR LAENTA AKAEQLIEGLITDGMKVKN AALDARTLGRPVDISAWZHA 240									
PP		*****									

Query	241	eqpaldfngmgptrlla 258									
Target	241	EQPALDFNGMGPMTRTLA 258									
PP		*****97									

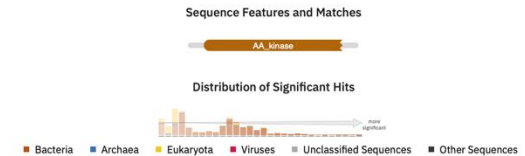
> Q3VV17		Acetylglutamate kinase						Shigella sonnei		3.3e-161	
> B2TWF4		Acetylglutamate kinase						Shigella boydii		4.9e-161	

Domain search for K22478 sequence



Search Details							Customise Table				
Target		Description					Species	E-value			
▼ Q87EL2		Acetylglutamate kinase					Xylella fastidiosa	5.2e-282			
Query		Target Envelope		Target Alignment						Domain E-value	
start	end	start	end	start	end	Bias	% Identity (count)	% Similarity (count)	Bit Score	Independent	Conditional
18	438	1	421	1	421	0.17	100.0 (421)	100.0 (421)	947.0	5.8e-282	8.8e-286
*****.....*											
Query	18	masakeisylkrfqlndakravvkvgavlrddvltssiflqevltipvlhagngldelvtgactkctmg					97				
Target	1	masakeisylkrfqlndakravvkvgavlrddvltssiflqevltipvlhagngldelvtgactkctmg					98				
PP		89)*****.....*									
*****.....*											
Query	98	rvtlptemalivkvfhntlnlealngaratsitggvfeshyldqetylvlgvisavnapiaslaasvpas					177				
Target	81	rvtlptemalivkvfhntlnlealngaratsitggvfeshyldqetylvlgvisavnapiaslaasvpas					160				
PP		81)*****.....*									
*****.....*											
Query	178	gttspgellinadvaanvlhlylpyksifltftgtllgdgkikinsltreyeklqqevygpklkiektlvld					257				
Target	161	gttspgellinadvaanvlhlylpyksifltftgtllgdgkikinsltreyeklqqevygpklkiektlvld					240				
PP		161)*****.....*									
*****.....*											
Query	258	rlilesevsrtedpealeknkgstltzgevratetwldtpslchliassrttlipshyettollayven					337				
Target	241	rlilesevsrtedpealeknkgstltzgevratetwldtpslchliassrttlipshyettollayven					320				
PP		241)RLLESVSRTEDPADLAKELTHKSGLTZGEVRATETWLDTPSLCHLIQSSEHRTLLPHYETTOLLAYVEN									
*****.....*											
Query	338	vraaviltnlgnpyldkdvldsddegglgrawsmmshrsnmmamyaasdydknmkshfeghl					417				
Target	321	vraaviltnlgnpyldkdvldsddegglgrawsmmshrsnmmamyaasdydknmkshfeghl					400				
PP		321)VRAAVILTNLGNPYLDKDVLDSDDEGLGRAWSMMSHRSNMMAMYAASDYDKNMKSHFEGHL									
*****.....*											
*****.....*											
Query	418	hfioiqevahvthoptltd					438				
Target	401	HFQIQOVCVAHQCTHOPTLTG					421				
PP		401)*****.....*									

Domain search for NC_012660.1_5881 ORF sequence



Search Details		Customise Table	
Target	Description	Species	E-value
> A0A3M5PWT5	Acetylglutamate kinase	<i>Pseudomonas marginalis</i>	6.4e-188
> A0A810CZJ5	Acetylglutamate kinase	<i>Pseudomonas tritici</i>	2.3e-187
> A0A3M3ECE1	Acetylglutamate kinase	<i>Pseudomonas corrugata</i>	4.1e-184
> A0A850HG25	Acetylglutamate kinase	<i>Pseudomonas</i> sp. B014	2.3e-183
> A0A0C5EDU4	Acetylglutamate kinase	<i>Pseudomonas</i> sp. MRSN 12121	1.3e-180
> A0A1H1SC34	Acetylglutamate kinase	<i>Pseudomonas asplenii</i>	2.4e-178
> A0A0T4UHQ6	Acetylglutamate kinase	<i>Pseudomonas</i> sp. T11014	1.5e-177

HMMSEARCH for K00930.hmm against ORF NC_012660.1_5881

```
Query: K00930 [M=400]
Scores for complete sequences (score includes all domains):
--- full sequence --- --- best 1 domain --- --dom-
E-value score bias E-value score bias exp N Sequence Description
1.9e-132 440.3 9.2 2.1e-132 440.1 9.2 1.0 1 NC_012660.1_5881 # 6543745 # 6544650 # 1 # ID=1_5881;partial
3.6e-42 143.0 0.0 4.5e-42 142.7 0.0 1.1 1 NC_012660.1_5758 # 6417456 # 6418757 # -1 # ID=1_5758;partia
5.6e-12 43.6 2.2 1.3e-08 32.5 0.3 2.1 2 NC_012660.1_5068 # 5663447 # 5664565 # -1 # ID=1_5068;partia
3.1e-10 37.8 4.3 6.7e-10 36.8 4.2 1.6 1 NC_012660.1_4649 # 5220972 # 5222213 # -1 # ID=1_4649;partia
2.5e-08 31.6 1.9 1.3e-06 25.9 0.3 2.1 2 NC_012660.1_4796 # 5371640 # 5372569 # 1 # ID=1_4796;partial
----- inclusion threshold -----
0.067 10.4 0.0 0.095 9.9 0.0 1.1 1 NC_012660.1_5317 # 5949314 # 5950234 # -1 # ID=1_5317;partia
```

Domain annotation for each sequence (and alignments):

```
>> NC_012660.1_5881 # 6543745 # 6544650 # 1 # ID=1_5881;partial=00;start_type=ATG;rbs_motif=GGAG/GAGG;rbs_spacer=5-10bp
# score bias c-Evalue i-Evalue hmmfrom hmm to alifrom ali to envfrom env to acc
1 ! 440.1 9.2 2.1e-135 2.1e-132 75 353 .. 11 297 .. 1 301 [. 0.96
```

Alignments for each domain:

== domain 1 score: 440.1 bits; conditional E-value: 2.1e-135

```
K00930 75 kaevLaealpyiqkfrgktvVvKyGgNnamkkaafardivLLklvgrlpVvVHGgGppinalllkrlgiesefvnlRvtdaetmevvemv 167
+avvL+ealpyi+++ gkt+V+KyGgNam+++elk fardivl+k+vg++pVvVHGgGpp1 +llkrl ies+f++G+Rvtda+tm+vvemv
NC_012660.1_5881 11 TAKVLSEALPYIRRYVGKTLVIKYGGNAMESDELKTGFARDIVLMKAVGINPVVHHGGGPQIGDLLKRLSIESHFIDGMRVTDATQMDVVEMV 103
89***** PP
```

```
K00930 168 LagkvnelVslinaaggkAvGlsqkDgnlltarkl.....dgeevdLGfVGevtvdpsslralldagliPViapiavdedgqvyni 250
L+g+vnk++V+lin++gg A+Gl+gkD+ l++a+kl + e +d+G+VGev +++++ l++ ++iPViapi+v++g++yNi
NC_012660.1_5881 104 LGGQVNKDIVNLINRHGGSAILTKDAELIRAKKLTvtrqtpemtQPEIIDIGQVGEVIGINTDLLNLLVKGDFIPVIAPIGVGANGESYNI 196
*****99998876543479***** PP
```

```
K00930 251 NADtaAgaAaalgaekLilltDvpGvldwdpgsliseltaaereliadgtiagGMipKveaclralleggvksvhiidgrvphslLleift 343
NAD +Ag++A+al+aekL+llt+++G+dd ++g+++ lt +++eliadgti gGM+pK++++l+a+ggv s i+dgrvp+++Lleift
NC_012660.1_5881 197 NADLVAGKVAEALKAELMLLTNIAGLMD---KEGKVLTLGLTTQVDELIDAGDTIYGGMPLKIRCALEAVQGGVSSLLDGRVPNAILLEIFT 287
*****.99***** PP
```

```
K00930 344 degiGTmitd 353
d g+GT+i +
NC_012660.1_5881 288 DTGVGTLSN 297
*****974 PP
```

HMMSEARCH for K22478.hmm against ORF NC_012660.1_5881

```
Query: K22478 [M=435]
Scores for complete sequences (score includes all domains):
--- full sequence --- --- best 1 domain --- --dom-
E-value score bias E-value score bias exp N Sequence Description
8.1e-36 122.0 2.2 4.9e-35 119.5 2.2 1.8 1 NC_012660.1_5881 # 6543745 # 6544650 # 1 # ID=1_5881;partial
1e-21 75.6 0.0 1.4e-21 75.1 0.0 1.1 1 NC_012660.1_5758 # 6417456 # 6418757 # -1 # ID=1_5758;partia
0.00051 17.2 0.0 0.0006 17.0 0.0 1.1 1 NC_012660.1_2072 # 2294602 # 2295048 # -1 # ID=1_2072;partia
0.0065 13.6 0.0 0.0068 13.5 0.0 1.1 1 NC_012660.1_3336 # 3774578 # 3775054 # 1 # ID=1_3336;partial
```

Domain annotation for each sequence (and alignments):

```
>> NC_012660.1_5881 # 6543745 # 6544650 # 1 # ID=1_5881;partial=00;start_type=ATG;rbs_motif=GGAG/GAGG;rbs_spacer=5-10bp
# score bias c-Evalue i-Evalue hmmfrom hmm to alifrom ali to envfrom env to acc
1 ! 119.5 2.2 3.2e-38 4.9e-35 32 287 .. 25 298 .. 12 300 .. 0.83
```

Alignments for each domain:

== domain 1 score: 119.5 bits; conditional E-value: 3.2e-38

```
K22478 32 dakrfavvkvggavLrd.dl.eaLassSflqqvgLtpivvhGagpqldeelaagiekktdvdlrvtspealairrvrfq.asnlkLveaLe 121
+ +++++k gg+ ++ +l +a + ++ vg+ p+vvhG+gppq+ + l+ +ie++ +dg+rvt+ +++++v v+ + n ++v ++
NC_012660.1_5881 25 YVGKTLVIKYGGNAMESdELKTGFARDIVLMKAVGINPVVHHGGGPQIGDLLKRLSIESHFIDGMRVTDATQMDVVEMVLGgQVNKDIVNLIN 117
56799*****98455245899*****9636699999988 PP
```

```
K22478 122 eegaratsits...gvfeaelldet.....lGlvgevesvelapieaslragsipviaslgeteegqilnvnadfaanelvevlq 199
++g a +t+ ++a+ l + +G vgev +++ ++ ++ + ipvia +g ++++ n+nad +a +++e+l+
NC_012660.1_5881 118 RHGGSAILTLGkdaELIRAKKLTvtrqtpemtqpeiidIGQVGEVIGINTDLLNLLVKGDFIPVIAPIGVGANGESYINADLVAGKVAEALK 210
8887776666411344444332111111122222338***** PP
```

```
K22478 200 pykiifltgtGGLldekgkvidsinlsteyehliaqpwweggmrlkleikdLLdklplessvsi..TspaeLakElfthkGsGTLvrrg 287
k++ lt+ Gl+d++gkv+ + ++ ++lia+ ++ ggm k+ + + + ss+ + p+ ++ E+ft +G GTL+ +
NC_012660.1_5881 211 AEKLMLLTNIAGLMDKEGKVLTLGLTTQQ-VDELIADGTIYGGMPLKIRCALEAVQGGV-GSSLLIdgRVPNAILLEIFTDTGVGTLSNR 298
*****9887.999*****99987777665.444444114599*****9865 PP
```